

A blurred background image of a business meeting. Several people in professional attire are gathered around a table. One person is holding a smartphone, another is holding a tablet displaying a diagram with circles and lines. There are also coffee cups on the table. The overall scene suggests a collaborative work environment.

CSE345

BPM

Introduction to Business Process Management

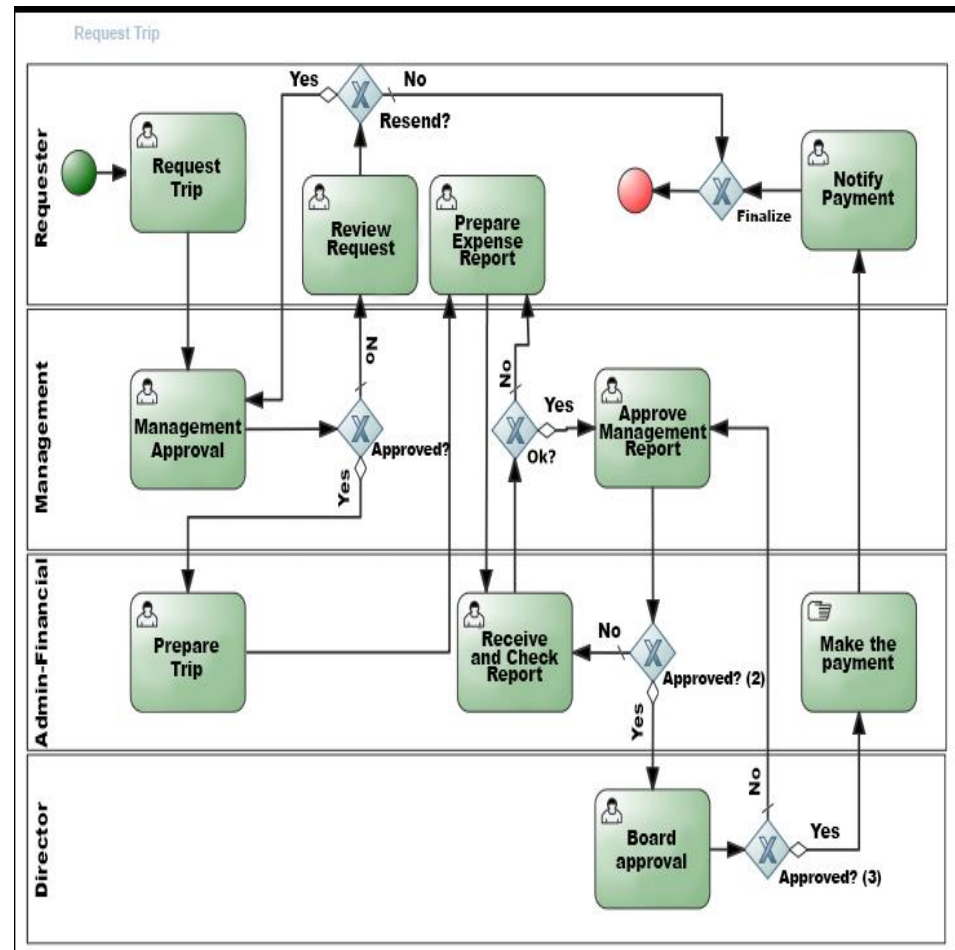
Dr. Islam El-Maddah

Why Business Process Management Matters Today

- Organizations suffer from delays, rework, and inefficiencies
 - Digital transformation exposes weak processes
 - Automation without BPM increases risk

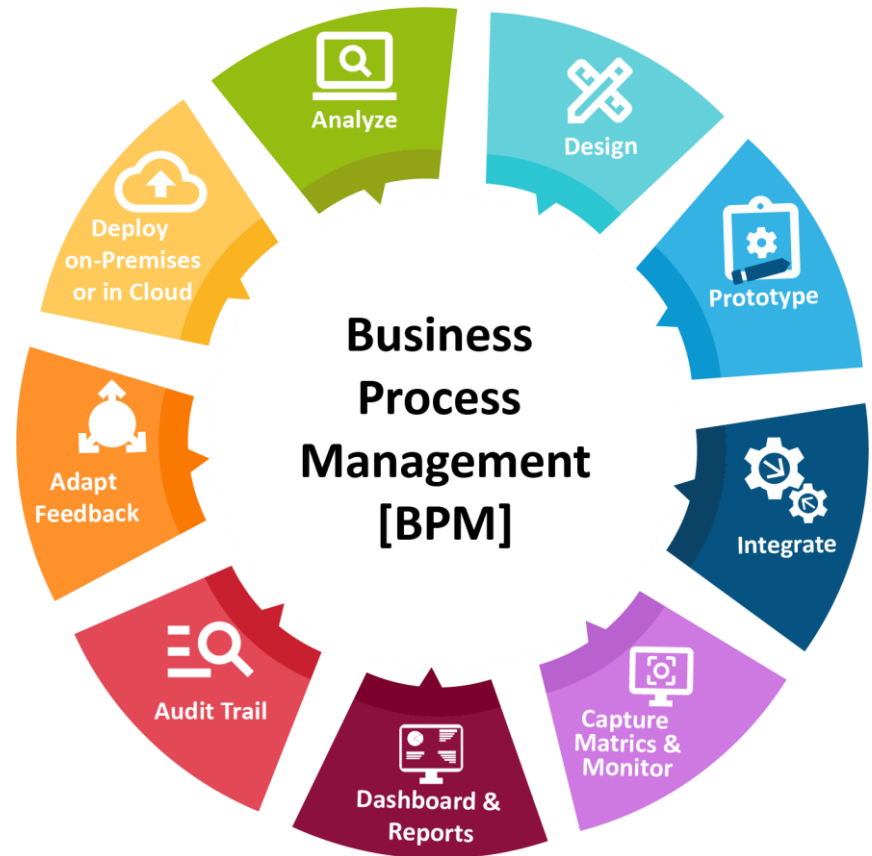
What Is a Business Process?

- A structured set of activities to achieve a business goal
 - Involves people, systems, data, and resources
 - Delivers value to customers



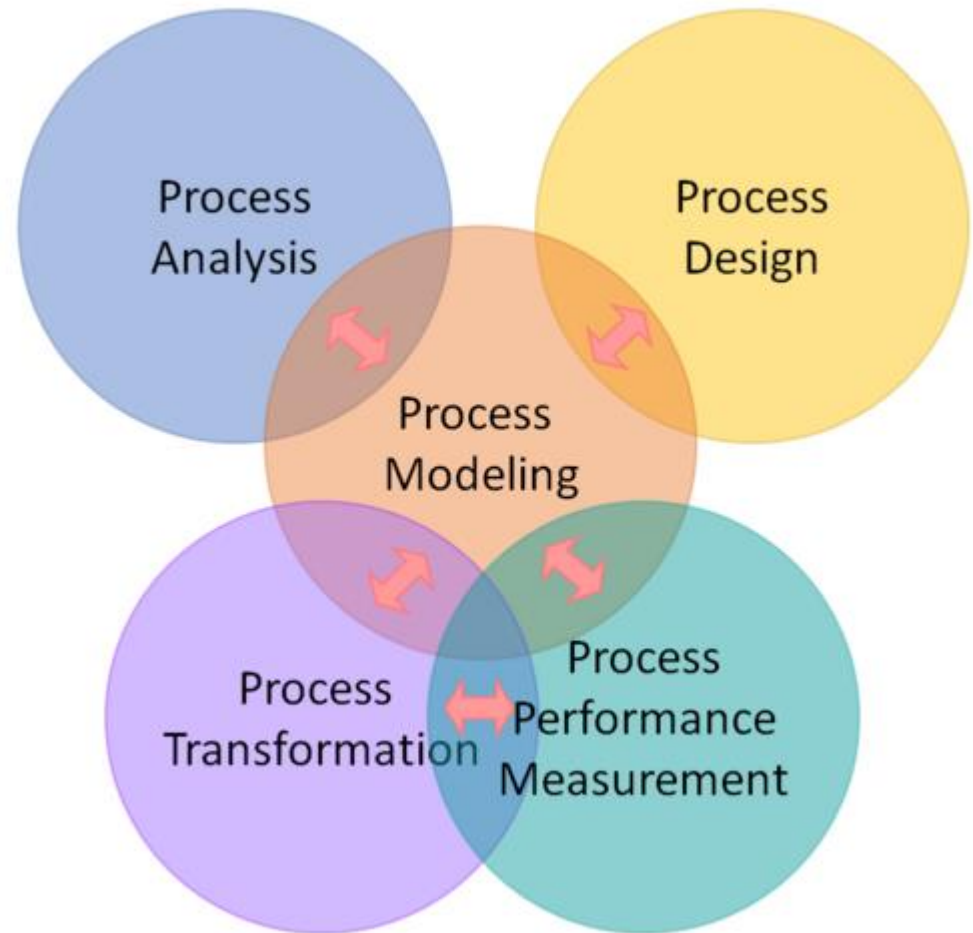
What Is Business Process Management (BPM)?

- A discipline for modeling, analyzing, and improving processes
 - Combines management and engineering thinking
 - Supports continuous improvement and automation



Why Model Business Processes?

- Make invisible work visible
 - Understand how work really happens
 - Enable analysis and improvement

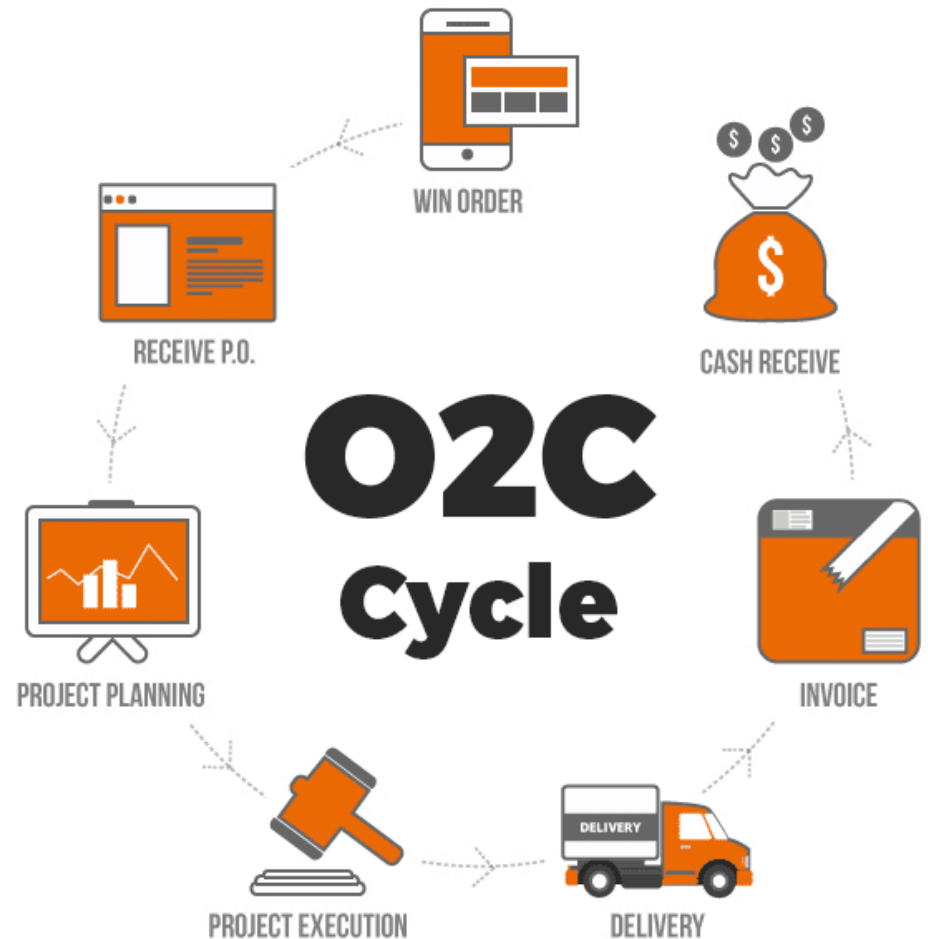


BPM Models vs UML Models

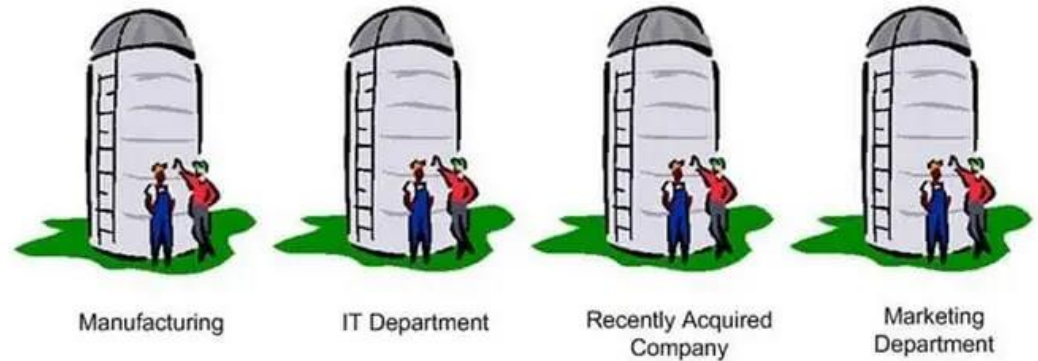
- BPM focuses on business workflows
 - UML focuses on software structure
 - BPM models are readable by non-technical users

Common End-to-End Business Processes

- Order-to-Cash (O2C)
 - Procure-to-Pay (P2P)
 - Claim-to-Settlement
 - Fault-to-Resolution



Functional Silos vs Process Thinking



- Functional silos optimize departments
 - Processes optimize customer outcomes
 - Hand-offs cause delays and errors

Why Process Problems Persist

- Processes are undocumented
 - Responsibilities are fragmented
 - Technology is added without redesign



Process Improvement Approaches



- Continuous Process Improvement (CPI)
 - Small, incremental improvements
 - Business Process Re-Engineering (BPR) for radical change

CPI vs BPR

- CPI improves existing processes
 - BPR redesigns processes from scratch
 - Both approaches are valid in different contexts

CPI vs BPR

Aspect	Small, Incremental Improvements (CPI / Continuous Improvement)	Business Process Re-Engineering (BPR) – Radical Change
Change Scope	Minor adjustments to existing processes	Complete redesign of end-to-end processes
Nature of Change	Gradual, evolutionary	Radical, revolutionary
Risk Level	Low risk	High risk
Cost	Relatively low	High initial investment
Time to Implement	Short to medium term	Medium to long term
Impact on Organization	Limited disruption	Major organizational disruption
Focus	Eliminating inefficiencies and waste	Achieving dramatic performance breakthroughs
Typical Improvements	5–20% improvement	50–90% improvement
Employee Involvement	High, continuous involvement	Selective involvement, top-down driven

CPI vs BPR

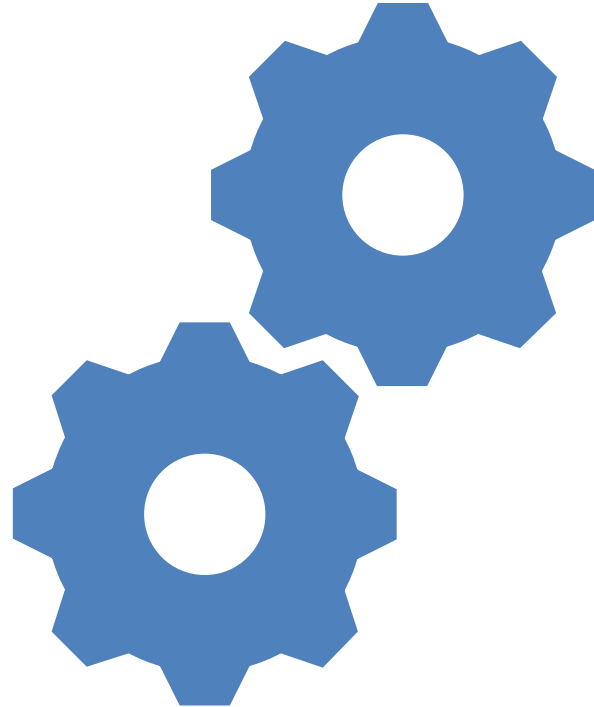
Aspect	Small, Incremental Improvements (CPI / Continuous Improvement)	Business Process Re-Engineering (BPR) – Radical Change
Cultural Impact	Supports continuous learning culture	Requires major cultural change
Technology Role	Supports existing processes	Often enables completely new ways of working
When to Use	Processes are basically sound but inefficient	Processes are fundamentally broken or obsolete
Example	Reducing approval steps in an existing workflow	Replacing manual claims handling with a digital self-service model

Classic Case: Ford Procurement

- Problem: slow and expensive procurement
 - Solution: process redesign before automation
 - Result: 75% reduction in staff

The BPM Lifecycle

- Define objectives
 - Model As-Is process
 - Design To-Be process
 - Implement and monitor



Value-Adding vs Non-Value-Adding Activities

- Value-adding: customer willing to pay
 - Business-value: control and compliance
 - Waste: delays, rework, waiting

Automation and BPM

- Automation follows process improvement
 - Bad processes become worse when automated
 - BPM guides effective use of IT and AI

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Modern BPM

- Process mining and analytics
 - Digital transformation enablement
 - AI-assisted decision support



Class Exercise

- Identify a real service process
 - Define goals and performance measures
 - Propose one improvement

Key Takeaways

- BPM focuses on value delivery
 - Processes must be understood before automation
 - Good processes enable digital success